



KAMBALA

Student Number: _____

Trial HSC - Task 3
August 2013

Mathematics Extension 2

General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black or blue pen
Black pen is preferred
- Board-approved calculators may be used
- Answer questions 1 – 10 on the multiple choice answer sheet provided.
- Answer questions 11 – 16 in the booklets provided.
Start each question in a new booklet.
- A table of standard integrals is provided at the back of this paper
- In Questions 11 – 16, show relevant mathematical reasoning and/or calculations

Total marks – 100

Section I – Pages 3 – 6

10 marks

- Attempt Questions 1 - 10
- Allow about 15 minutes for this section

Section II – Pages 7 – 14

90 marks

- Attempt Questions 11 - 16
- Allow about 2 hours 45 minutes for this section

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Section I

10 Marks

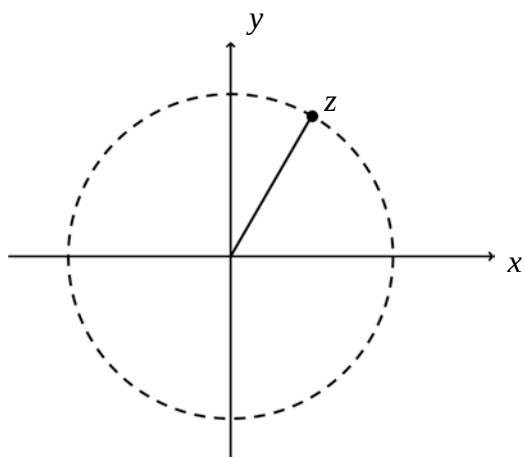
Attempt Questions 1 – 10

Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1 – 10.

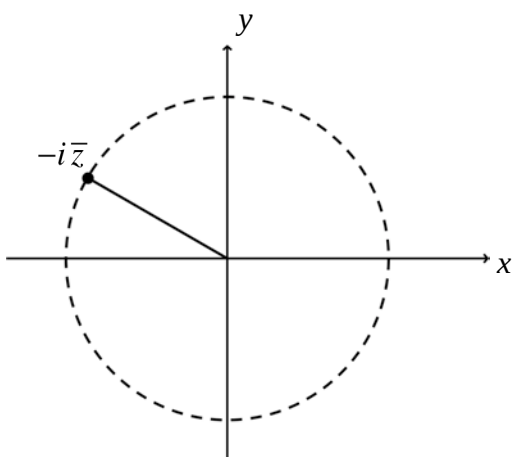
- 1 Let $z = 3 + 2i$ and $w = 2 - i$.
What is the value of $\bar{z} + 3w$?
- A. $9 + i$ B. $9 - i$ C. $9 - 3i$ D. $9 - 5i$
- 2 The equation $x^3 + 2xy - y^3 - 2 = 0$ defines y implicitly as a function of x .
What is the value of $\frac{dy}{dx}$ at the point $(1, 1)$?
- A. 5 B. 1 C. $\frac{1}{5}$ D. -1
- 3 The equation $3x^3 + 2x^2 - 4x + 1 = 0$ has roots α , β and γ .
What is the value of $\frac{1}{\alpha^3\beta^3\gamma^3}$?
- A. 27 B. -27 C. $\frac{1}{27}$ D. $-\frac{1}{27}$
- 4 What is the eccentricity of the hyperbola $\frac{x^2}{6} - \frac{y^2}{10} = 1$?
- A. $\frac{2\sqrt{10}}{5}$ B. $\frac{2\sqrt{6}}{3}$ C. $\frac{2\sqrt{3}}{3}$ D. $\frac{\sqrt{6}}{3}$
- 5 The polynomial $P(x) = 0$ has real coefficients. Three of the roots of $P(x)$ are $x = 4$, $x = 2 - i$ and $x = 1 + i$. What is the minimum degree of $P(x)$?
- A. 3 B. 4 C. 5 D. 6

- 6 The complex number z is shown in the Argand diagram below.

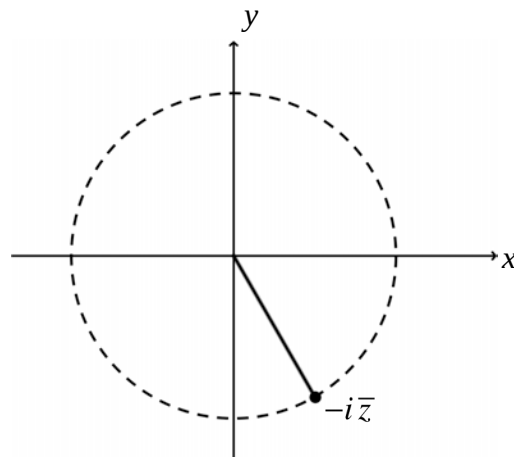


Which of the following best represents $-i\bar{z}$?

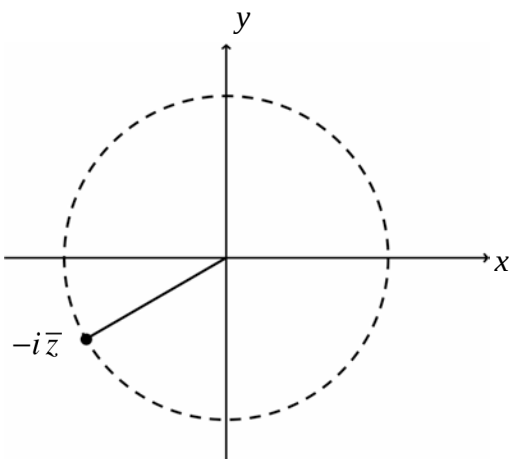
A.



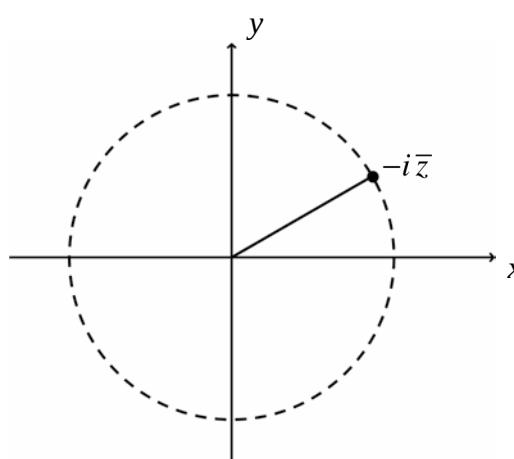
B.



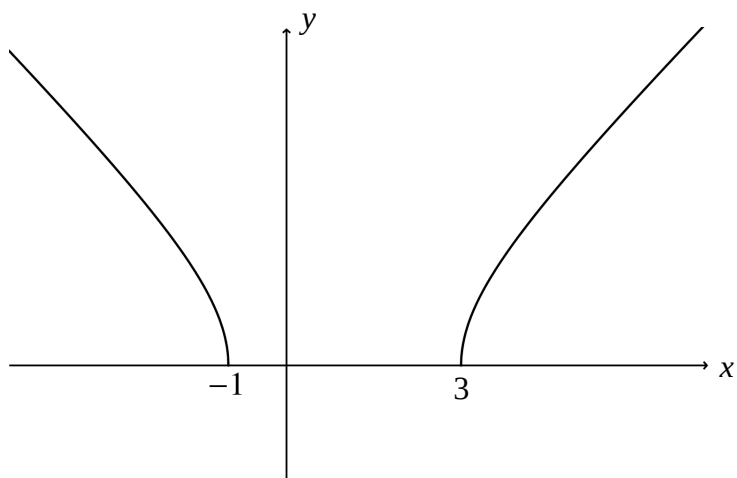
C.



D.

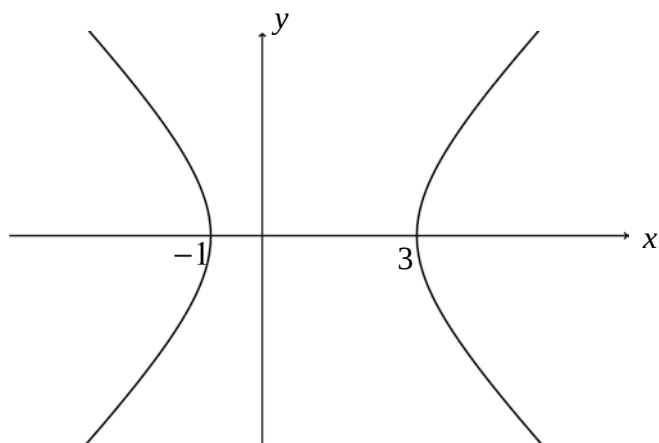


- 7 The graph of $y = \sqrt{f(x)}$ is shown below.

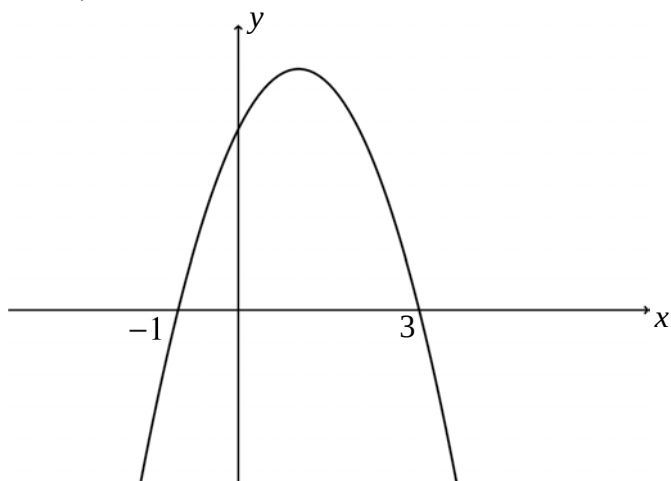


A possible graph of the function $y = f(x)$ is:

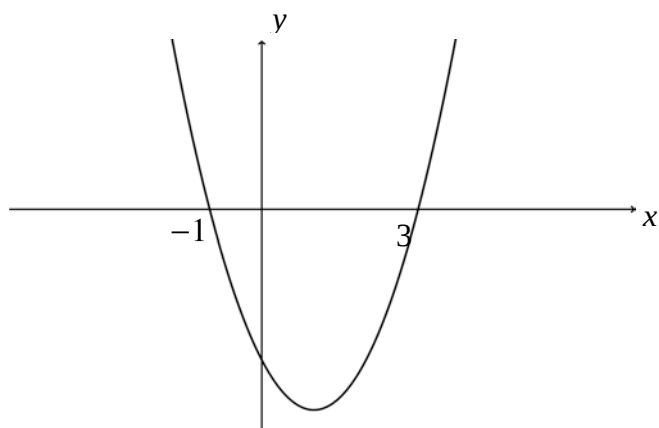
A.



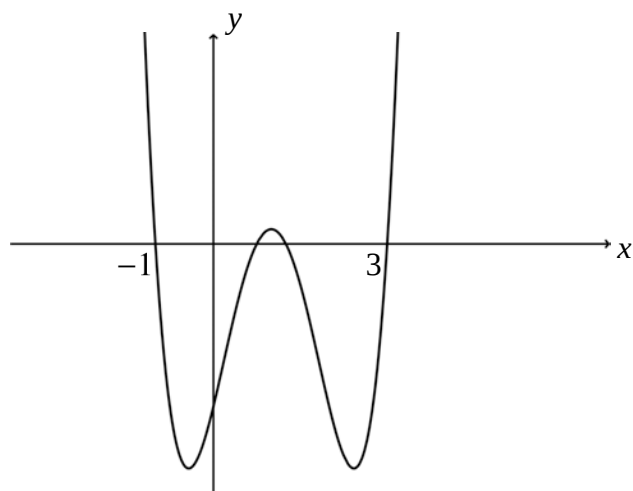
B.



C.



D.



- 8 If ω is an imaginary cube root of unity, then $(1 + \omega - \omega^2)^7$ is equal to:
- A. $-128\omega^2$ B. $128\omega^2$ C. -128ω D. 128ω
- 9 Let $y = \cos^{-1} e^x$. An expression for $\frac{dy}{dx}$ is given by:
- A. $-\tan y$ B. $-\cot y$ C. $-\operatorname{cosec} y$ D. $-\sec y$
- 10 What are all the values of k for which the graph of $y = 2x^3 - 6x^2 + k$ will have three distinct x -intercepts?
- A. $k > 0$ B. $k < 8$ C. $k = 0, 8$ D. $0 < k < 8$

End of Section I

Section II**60 Marks****Attempt Questions 11 – 16****Allow about 2 hours 45 minutes for this section**

Answer each question in the booklets provided. Start each question in a new booklet.
In Questions 11 – 16, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet.

(a) Express $\frac{2\sqrt{3}+i}{\sqrt{3}-i}$ in the form $x + iy$, where x and y are real. 2

(b) Shade the region on the Argand diagram where the two inequalities 2

$$|z+1| \leq 1 \quad \text{and} \quad |z-2i| \geq 2$$

both hold.

(c) Given $z_1 = i\sqrt{2}$ and $z_2 = \frac{2}{1-i}$

(i) Express z_1 and z_2 in modulus-argument form. 2

(ii) If $z_1 = wz_2$, find w in modulus-argument form. 1

(iii) On an Argand diagram, plot the points P , Q and R , where P represents z_1 , Q represents z_2 and R represents $(z_1 + z_2)$. 2

(iv) Show that $\text{Arg}(z_1 + z_2) = \frac{3\pi}{8}$ and hence find the exact value of $\tan \frac{3\pi}{8}$. 2

(d) Given that $2^{n+4} > (n+4)^2$ for all integral $n \geq 1$, show that $2^{3(a+2)} > 9(a+2)^2$. 2

(e) Solve the equation $x^3 - 8x^2 - 5x + 84 = 0$, given that one of the roots is equal to the sum of the other two roots. 2

Question 12 (15 marks) Use a SEPARATE writing booklet.

(a) Find $\int \frac{dx}{x^2 - 2x + 5}$. 2

(b) Find $\int \ln x \, dx$. 2

(c) Using the substitution $u = e^x$ and partial fractions, find $\int \frac{1}{e^x + 1} \, dx$. 3

(d) Prove by the process of Mathematical Induction that $(1+x)^n - nx - 1$ is divisible by x^2 for all integral $n \geq 2$. 3

(e) Given that $f(x) = |x - 2| - 2$, sketch the graphs of the following showing the x - and y -intercepts. Use separate axes for each graph.

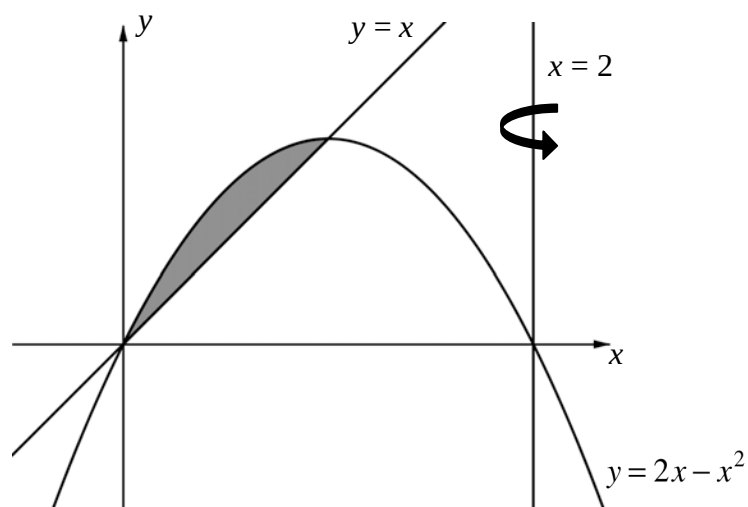
(i) $y = f(x)$ 1

(ii) $y = [f(x)]^2$ 2

(iii) $y = \ln[f(x)]$ 2

Question 13 (15 marks) Use a SEPARATE writing booklet.

- (a) The area between the two curves $y = 2x - x^2$ and $y = x$ is rotated about the line $x = 2$.



Using the method of cylindrical shells, calculate the exact volume of the solid of revolution formed.

3

- (b) (i) Find the centre and radius of the circle $x^2 + y^2 + 4x = 0$.
 (ii) Show that the line $y = mx + b$ will be a tangent to the circle if

1

2

$$4(mb + 1) = b^2$$

- (iii) P is a point whose coordinates are $(k, 0)$.
 If P lies on the line $y = mx + b$ and is exterior to the circle, find possible values for k if the two tangents from P to the circle are perpendicular.

2

- (c) (i) Show that $\int_0^a f(x) dx = \int_0^a f(a-x) dx$.

1

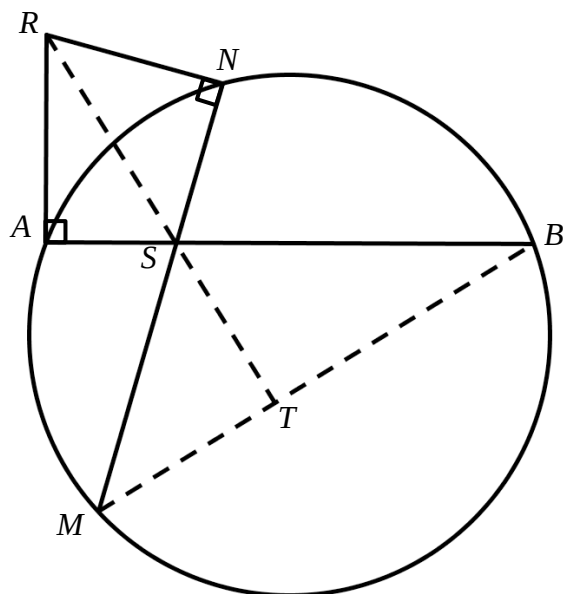
- (ii) Hence evaluate $\int_0^1 x^2 \sqrt{1-x} dx$.

3

Question 13 continues on page 10

Question 13 (continued)

- (d) AB and MN are chords of a circle that intersect at S . R is a point external to the circle such that RA is perpendicular to the chord AB and RN is perpendicular to the chord MN . The line RS is produced to T , a point lying on MB .



Copy or trace this diagram into your answer booklet.

Prove that RT is perpendicular to MB .

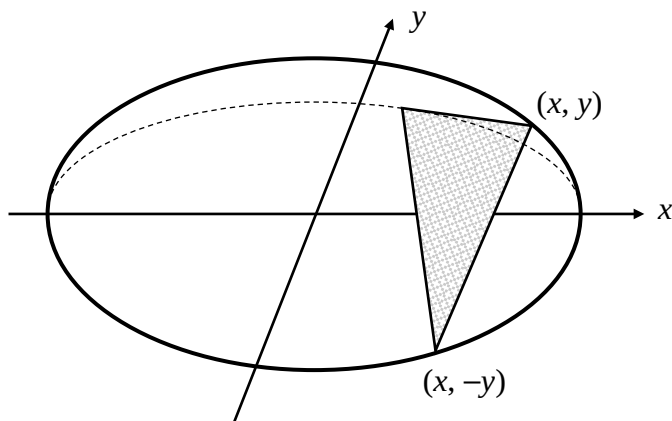
3

End of Question 13

Question 14 (15 marks) Use a SEPARATE writing booklet.

- (a) The base of a certain solid is in the shape of an ellipse with equation $\frac{x^2}{16} + \frac{y^2}{9} = 1$.

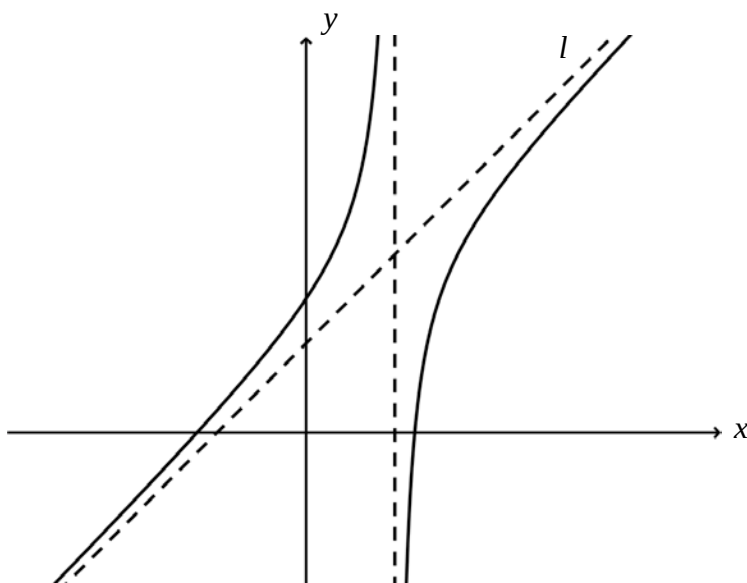
Sections parallel to the y -axis are equilateral triangles, with one side sitting in the base of the solid, as shown in the diagram below.



Find the exact volume of the shape.

4

- (b) The diagram shows the graph of $y = \frac{x^2 - 6}{x - 2}$. The line l is an asymptote.



- (i) Use the above graph to draw a one-third page sketch of the graph $y = \frac{x - 2}{x^2 - 6}$. 2
- (ii) By writing $\frac{x^2 - 6}{x - 2}$ in the form $mx + b + \frac{a}{x - 2}$, find the equation of the line l . 2

Question 14 continues on page 12

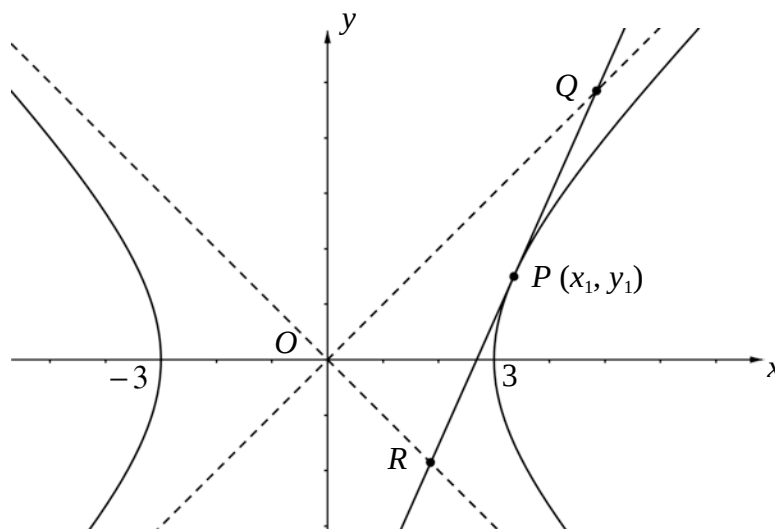
Question 14 (continued)

- (c) (i) By considering the expansion of $(\cos\theta + i\sin\theta)^3$ and de Moivre's theorem, 1
show that $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$.
- (ii) Deduce that $8x^3 - 6x - 1 = 0$ has solutions $x = \cos\theta$, where $\cos 3\theta = \frac{1}{2}$. 2
- (iii) Find the roots of $8x^3 - 6x - 1 = 0$ in the form $\cos\theta$. 2
- (iv) Hence evaluate $\cos\left(\frac{\pi}{9}\right)\cos\left(\frac{2\pi}{9}\right)\cos\left(\frac{4\pi}{9}\right)$. 2

End of Question 14

Question 15 (15 marks) Use a SEPARATE writing booklet.

- (a) The hyperbola with equation $x^2 - y^2 = 9$ is shown in the diagram below. The point $P(x_1, y_1)$ lies on the hyperbola. The tangent to the hyperbola at P intersects the asymptotes of the hyperbola at points Q and R .



- (i) Show that e , the eccentricity of the hyperbola, is equal to $\sqrt{2}$. 1
- (ii) Determine the coordinates of the foci, equations of the directrices and the equations of the asymptotes. 3
- (iii) Show that the equation of the tangent at P is 2

$$yy_1 = xx_1 - 9$$

- (iv) Prove that the area of triangle QOR is constant, where O is the origin. 3

- (b) The cubic function $y = x^3 - px + q$ has two turning points.

- (i) Show that $p > 0$. 1
- (ii) The line $y = k$ intersects this cubic in three distinct points. 3

Show that $q - \frac{2p}{3}\sqrt{\frac{p}{3}} < k < q + \frac{2p}{3}\sqrt{\frac{p}{3}}$.

- (c) The equation $x^3 - x^2 - 3x + 5 = 0$ has roots α , β and γ . Find the equation whose roots are $(2\alpha + \beta + \gamma)$, $(\alpha + 2\beta + \gamma)$ and $(\alpha + \beta + 2\gamma)$. 2

Question 16 (15 marks) Use a SEPARATE writing booklet.

- (a) $T_1, T_2, T_3, \dots$ are terms of an arithmetic sequence with common difference d . All terms in the sequence are positive.

(i) Show that $\frac{1}{\sqrt{T_{n-1}} + \sqrt{T_n}} = \frac{\sqrt{T_n} - \sqrt{T_{n-1}}}{d}$ for $n = 2, 3, 4, \dots$ 2

- (ii) Hence or otherwise, show that 2

$$\frac{1}{\sqrt{T_1} + \sqrt{T_2}} + \frac{1}{\sqrt{T_2} + \sqrt{T_3}} + \dots + \frac{1}{\sqrt{T_{n-1}} + \sqrt{T_n}} = \frac{n-1}{\sqrt{T_1} + \sqrt{T_n}}$$

for $n = 2, 3, 4, \dots$

- (b) The equations of two conics are $3x^2 + 4y^2 = 48$ and $3x^2 - y^2 = 3$.

- (i) Show that these two conics have the same pair of foci. 1

- (ii) The point $(4\cos\theta, 2\sqrt{3}\sin\theta)$ lies on the ellipse for **all** values of θ . 2
Find the four values of θ for which this point also lies on the other conic.

- (iii) Show that the two conics intersect at right angles. 3

(c) (i) Show that $\left(1-x^2\right)^{\frac{n-3}{2}} - \left(1-x^2\right)^{\frac{n-1}{2}} = x^2\left(1-x^2\right)^{\frac{n-3}{2}}$ 1

(ii) Let $I_n = \int_0^1 \left(1-x^2\right)^{\frac{n-1}{2}} dx$ where $n = 1, 2, 3, \dots$ 3

Show that $nI_n = (n-1)I_{n-2}$ for $n = 2, 3, 4, \dots$

- (iii) Evaluate I_5 . 1

End of Paper

STANDARD INTEGRALS

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \quad \text{if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax \, dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax \, dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax \, dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax \, dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

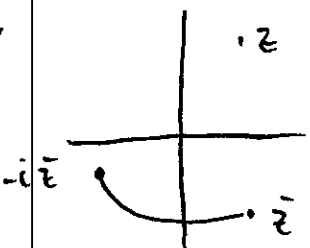
$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE: $\ln x = \log_e x, \quad x > 0$

Qn	Solutions	Marks	Comments & Criteria
1	$z = 3 + 2i$ $w = 2 - i$ $\bar{z} = 3 - 2i$ $3w = 6 - 3i$ $\therefore \bar{z} + 3w = 3 - 2i + 6 - 3i$ $= 9 - 5i$	<u>D</u>	
2.	$x^3 + 2xy - y^3 - 2 = 0$ $3x^2 + 2y + 2x \frac{dy}{dx} - 3y^2 \frac{dy}{dx} = 0$ $(2x - 3y^2) \frac{dy}{dx} = -3x^2 - 2y$ $\frac{dy}{dx} = \frac{-3x^2 - 2y}{(2x - 3y^2)}$ At (1, 1) $\frac{dy}{dx} = \frac{-3 - 2}{2 - 3}$ $= \frac{-5}{-1}$ $= 5$	<u>A</u>	
3.	$3x^3 + 2x^2 - 4x + 1 = 0$ $2/18 = -\frac{1}{3}$ $(2/18)^3 = -\frac{1}{27}$ $\therefore \frac{1}{2^3/18^3} = -27$	<u>B</u>	
4.	$\frac{x^2}{6} - \frac{y^2}{10} = 1$ $a^2 = 6$ $b^2 = 10$ $b^2 = a^2(e^2 - 1)$ $10 = 6(e^2 - 1)$		

$$10 = 6e^2 - 6 \rightarrow 16 = 6e^2 \rightarrow e^2 = \frac{16}{6} \rightarrow e = \frac{4}{\sqrt{6}} = \frac{4\sqrt{6}}{6} = \frac{2\sqrt{6}}{3}$$

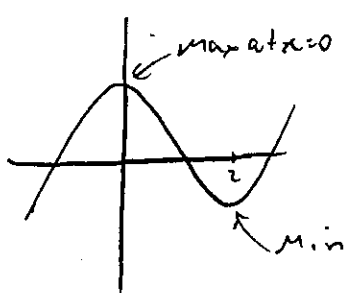
B

Qn	Solutions	Marks	Comments & Criteria
5.	Roots $x = 4, x = 2-i, x = 1+i$ As $P(x)$ has real coefficients, Complex roots occur in conjugate pairs $\therefore$ Other roots are $x = 2+i, x = 1-i$ $\therefore$ Min degree of $P(x)$ is 5.		<u>C</u>
6.	 $-i\bar{z}$ is a rotation through $\frac{\pi}{2}$ in a clockwise direction $\therefore$ <u>C</u>		
7.	Both A and C are possible, but A is <u>not</u> a function (fails vertical line test) $\therefore f(x)$ must be C	<u>C</u>	
8.	ω is a ^{cube} nth root of unity so a soln to $z^3 = 1$ i.e. $z^3 - 1 = 0$ Other roots are 1 and ω^2 $1 + \omega + \omega^2 = 0$ (sum of roots) $\therefore 1 + \omega = -\omega^2$ $(1 + \omega - \omega^2)^7 = (-2\omega^2)^7$ $= -128 \omega^{14}$		

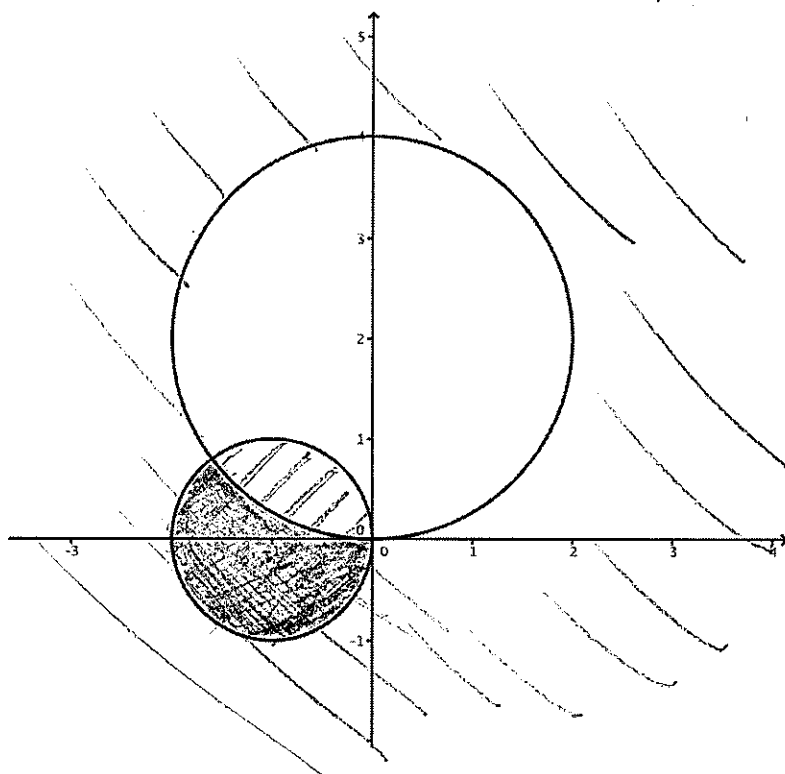
$$= -128 \omega^3 \cdot \omega^3 \cdot \omega^3 \cdot \omega^3 \cdot \omega^2$$

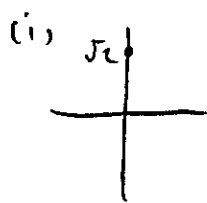
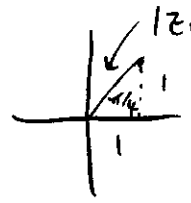
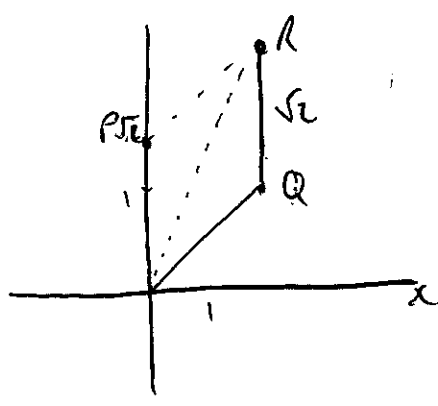
$$= -128 \omega^2 \text{ as } \omega^3 = 1$$

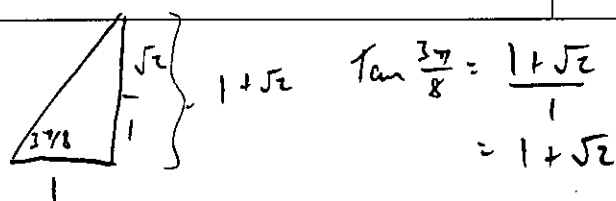
A
✓

Qn	Solutions	Marks	Comments & Criteria
9.	$y = \cos^{-1} e^x$ $\therefore e^x = \cos y$ $\frac{dy}{dx} (-\sin y) = e^x$ $= \cos y$ $\therefore \frac{dy}{dx} = \frac{\cos y}{-\sin y}$ $= -\cot y$	<u>B</u>	
10.	$y = 2x^3 - 6x^2 + k$ $y' = 6x^2 - 12x$ $= 6x(x-2) \therefore x = 0, 2 \text{ are stationary}$ Graph  At $x = 0$ $0 = 0 - 0 + k$ $k = 0$ At $x = 2$ $0 = 16 - 24 + k$ $k = 8$ $\therefore$ If k lies between 0 and 8 there are 3 roots $0 < k < 8$	<u>D</u>	

Qn	Solutions	Marks	Comments & Criteria
11	(a) $\frac{2\sqrt{3} + i}{\sqrt{3} - i} \times \frac{\sqrt{3} + i}{\sqrt{3} + i}$ $= \frac{(2\sqrt{3} + i)(\sqrt{3} + i)}{3 + 1}$ $= \frac{6 + 2\sqrt{3}i + \sqrt{3}i - 1}{4}$ $= \frac{5 + 3\sqrt{3}i}{4}$ $= \frac{5}{4} + \frac{3\sqrt{3}}{4}i$		1 correct conjugate 1 simplified answer
(b)	$z + 1 \leq 1$ $z - (-1 + 0i) \leq 1$ Circle ctr $(-1, 0)$, Radius 1 unit	$z - 2i \geq 2$ $z - (0 + 2i) \geq 2$ Circle Ctr $(0, 2)$ Radius 2 units	1 diagrams 1 regions correct



Qn	Solutions	Marks	Comments & Criteria
11	(c) $z_1 = i\sqrt{2}$ $z_2 = \frac{z}{1-i} \times \frac{1+i}{1+i}$ (i)  $z_1 = \sqrt{2} \operatorname{cis} \frac{\pi}{2}$ $z_2 = \frac{z+zi}{2}$ $= 1+i$ $z_2 = \sqrt{2}$  $z_2 = \sqrt{2} \operatorname{cis} \frac{\pi}{4}$ (ii) $z_1 = wz_2$ i.e. $w = \frac{z_1}{z_2}$ $= \frac{\sqrt{2}}{\sqrt{2}} \operatorname{cis} \left(\frac{\pi}{2} - \frac{\pi}{4} \right)$ $= \operatorname{cis} \frac{\pi}{4}$ (iii)  (iv) $\angle POQ = \frac{\pi}{2} - \frac{\pi}{4}$ $= \frac{\pi}{4}$ $\angle QOR$ bisects $\angle POQ = \frac{\pi}{8}$ as $OPRQ$ is a rhombus $\therefore \angle ROx = \frac{\pi}{4} + \frac{\pi}{8} = \frac{3\pi}{8}$		1 for z_1 1 for z_2 1 answer 1 for $\angle POQ$ 1 for R based on $\angle POQ$ 1 for showing



1 for correct value based on diagram

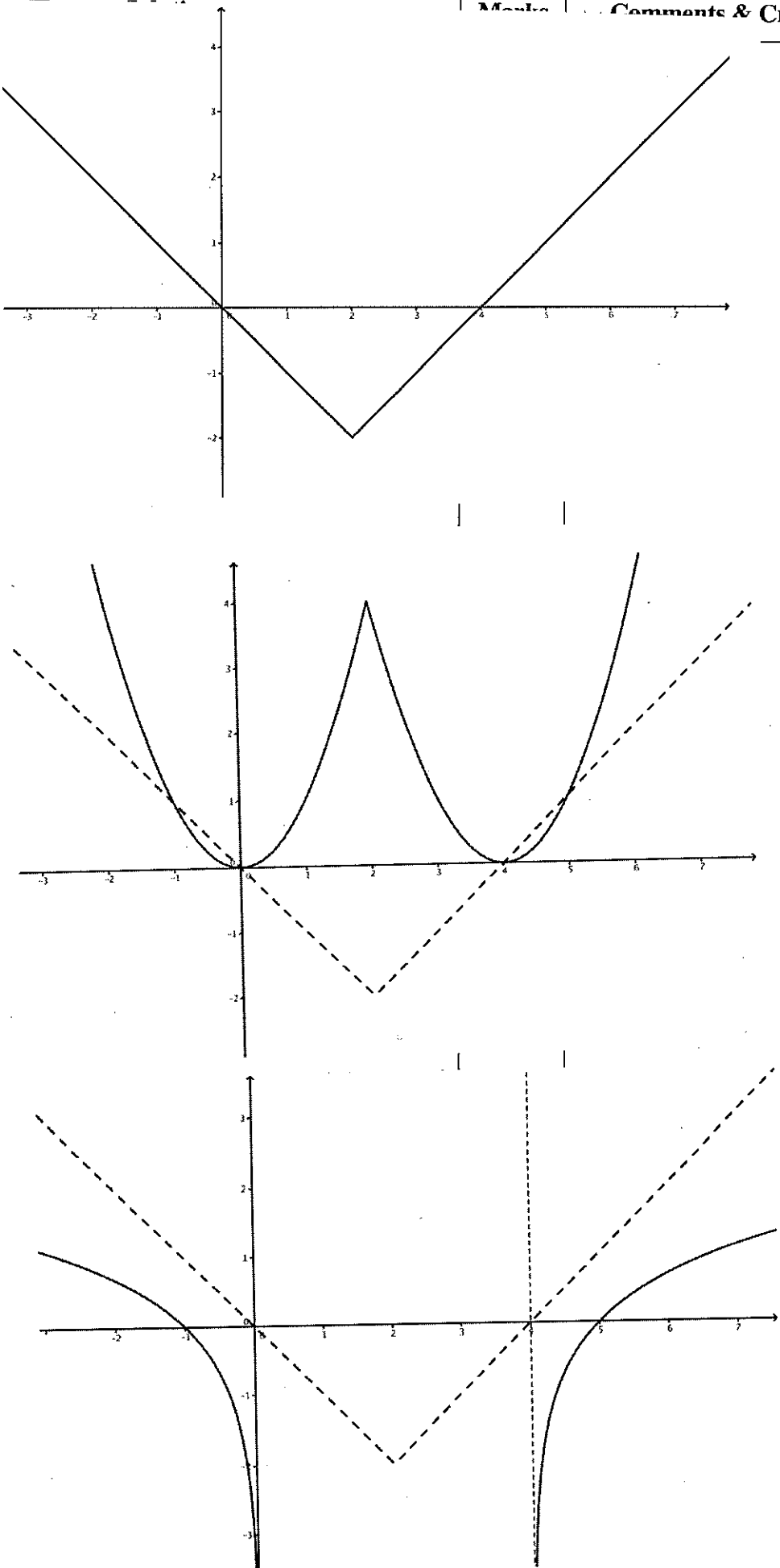
Qn	Solutions	Marks	Comments & Criteria
11	(d) $2^{n+4} > (1+4)^4 \quad n \geq 1$ To show $2^{3(a+2)} > 9(a+2)^2$ i.e. $2^{3a+6} > 3^2(a+2)^2$ $2^{3a+6} > (3a+2)^2$ Using result given $n+k = 3a+6$ $n = 3a+2$ $\therefore$ If you let $n = 3a+2$ Then $2^{3(a+2)} > 9(a+2)^2$ (e) $x^3 - 8x^2 - 5x + 84 = 0$ Let roots be $\alpha, \beta, (\alpha+\beta)$ Sum of roots: $\alpha + \beta + (\alpha + \beta) = \frac{-(-8)}{1}$ $2\alpha + 2\beta = 8$ $\alpha + \beta = 4 \Rightarrow \alpha = (4 - \beta)$ Product $\alpha\beta(\alpha+\beta) = \frac{-84}{1}$ $(4 - \beta)\beta(4 - \beta + \beta) = -84$ $(4\beta - \beta^2)4 = -84$ $16\beta - 4\beta^2 = 21$ $\beta^2 - 4\beta - 21 = 0$ $(\beta - 7)(\beta + 3) = 0$ $\therefore \beta = 7 \text{ or } -3$		1 for progress 1 for substitution 1 for sum/product or progress

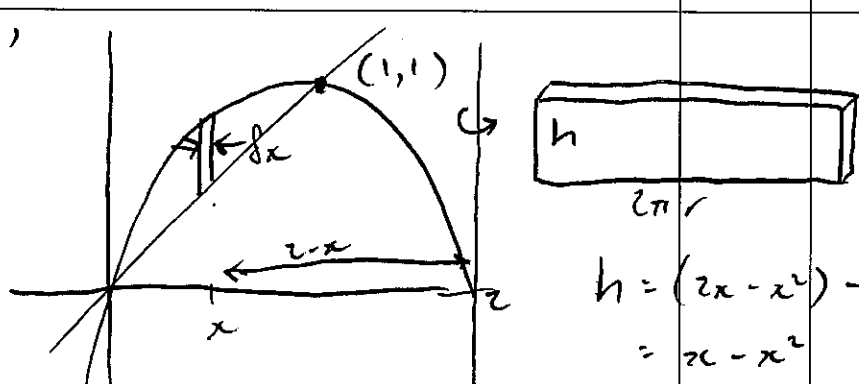
$\therefore$ Roots $x = -3, 7, -3+7$

i.e. $x = -3, 4, 7$

1 correct solve

Qn	Solutions	Marks	Comments & Criteria
12	(a) $\int \frac{dx}{x^2 - 2x + 5}$ $= \int \frac{dx}{x^2 - 2x + 1 + 4}$ $= \int \frac{dx}{(x-1)^2 + 2^2}$ $= \int \frac{dx}{2^2 + (x-1)^2}$ $= \frac{1}{2} \tan^{-1} \left(\frac{x-1}{2} \right) + c$ (b) $\int \ln x \, dx$ $= \int 1 \cdot \ln x \, dx$ $= x \ln x - \int \frac{1}{x} \cdot x \, dx$ $= x \ln x - \int 1 \cdot dx$ $= x \ln x - x + c$ (c) $\int \frac{1}{e^x + 1} \, dx$ $= \int \frac{1}{u+1} \cdot \frac{du}{u}$ $= \int \frac{1}{u(u+1)} \, du$ Now $\frac{1}{u(u+1)} = \frac{A}{u} + \frac{B}{u+1}$ $1 = A(u+1) + Bu$ Let $u=0 \Rightarrow A=1$ $u=-1 \Rightarrow 1=-B$ $B=-1$	$u = \ln x$ $u' = \frac{1}{x}$ $v = x$ $v' = 1$ Let $u = e^x$ $\frac{du}{dx} = e^x$ $du = e^x dx$ $dx = \frac{du}{e^x}$	1 completing square 1 correct integral 1 for parts set up 1 for correct answer 1 correct substituti. set up 1 for partial fractions

Qn	Marks	Comments & Criteria
		1 1 for shape 1 for point 1 asympt 1 correct curve

Qn	Solutions	Marks	Comments & Criteria
13	(a)  $\delta V = 2\pi r h \delta x$ $= 2\pi (2-x)(x-x^2) \delta x$ $= 2\pi (2x - 2x^2 - x^2 + x^3) \delta x$ $= 2\pi (2x - 3x^2 + x^3) \delta x$ $V = \sum_{x=0}^1 2\pi (2x - 3x^2 + x^3) \delta x$ $= \lim_{\delta x \rightarrow 0} \sum_{x=0}^1 2\pi (2x - 3x^2 + x^3) \delta x$ $V = 2\pi \int_0^1 (x^3 - 3x^2 + 2x) dx$ $= 2\pi \left[\frac{x^4}{4} - x^3 + x^2 \right]_0^1$ $= 2\pi \left[\left(\frac{1}{4} - 1 + 1 \right) - (0 - 0 + 0) \right]$ $= 2\pi \times \frac{1}{4}$ $= \frac{\pi}{2} \text{ m}^2$ (b) (i), $x^2 + y^2 + kx = 0$ $x^2 + kx + 4 + y^2 = 4$ $(x+2)^2 + y^2 = 4$ $\therefore C(-2, 0)$ Radius 2 units	$h = (2x - x^2) - x$ $= x - x^2$ $r = 2 - x$ 1 for correct h, r 1 for integral 1 correct answer	

Qn	Solutions	Marks	Comments & Criteria
	(b)(ii) $y = mx + b$ $(x+2)^2 + y^2 = 4$ i.e. $(x+2)^2 + (mx+b)^2 = 4$ $x^2 + 4x + 4 + m^2x^2 + 2mbx + b^2 - 4 = 0$ $(m^2+1)x^2 + (2mb+4)x + b^2 = 0$ Tangent if $\Delta = 0$ i.e. $b^2 - 4ac = 0$ $(2mb+4)^2 - 4(m^2+1)(b^2) = 0$ $4m^2b^2 + 16mb + 16 - 4m^2b^2 - 4b^2 = 0$ $16mb + 16 = 4b^2$ $4mb + 4 = b^2$ $4(mb+1) = b^2$ as req'd (iii) $P(k, 0)$ $y = mx + b$ At P $0 = mk + b$ $b = -mk$ Sub into result from (ii) $4[m(-mk) + 1] = (-mk)^2$ $-4m^2k + 4 = m^2k^2$ $m^2k^2 + 4m^2k - 4 = 0$ $(k^2 + 4k)m^2 - 4 = 0$ Two possible values of m if tangents are perpendicular i.e. $m_1 m_2 = -1 \Rightarrow$ product of roots $= -1$ $\therefore \frac{-4}{k^2 + 4k} = -1$ $k^2 + 4k = 4$		1 for gradient 1 for discriminant vol. or similar 1 for proper

$$k^2 + 4k + 4 = 4 + 4$$

$$(k+2)^2 = 8$$

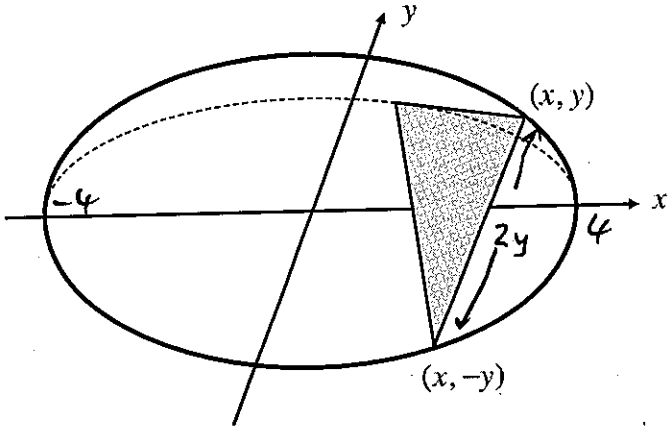
$$k+2 = \pm 2\sqrt{2}$$

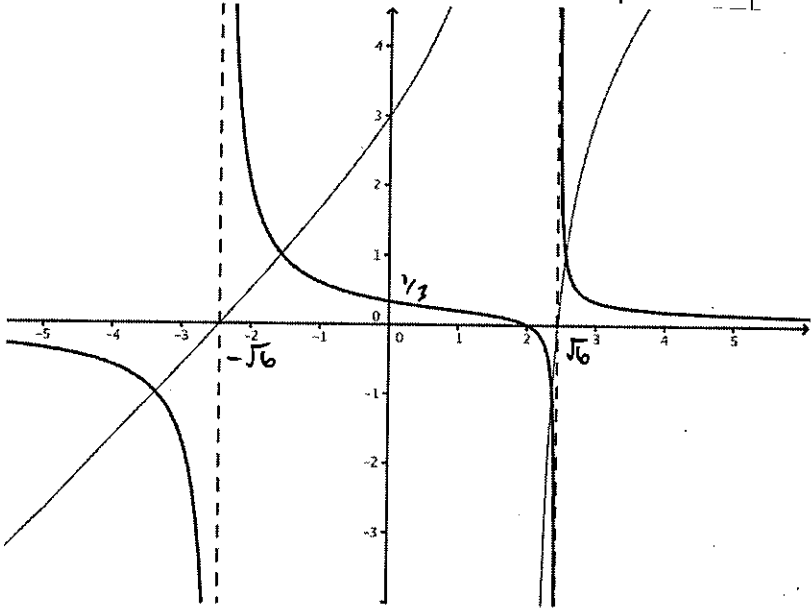
$$k = -2 \pm 2\sqrt{2}$$

1 for 1st value

Qn	Solutions	Marks	Comments & Criteria
13	(c) (i) $\int_0^a f(x) dx$ Let $u = a - x$ $x = a - u$ When $x = a$, $u = 0$ $x = 0$, $u = a$ $= \int_a^0 f(a-u) (-du)$ $= \int_0^a f(a-u) du$ $= \int_0^a f(a-x) dx$ as integral is independent of variable (ii) $\int_0^1 x^2 \sqrt{1-x} dx$ $= \int_0^1 (1-x)^2 \sqrt{1-(1-x)} dx$ $= \int_0^1 (1-x)^2 \sqrt{x} dx$ $= \int_0^1 (1 - 2x + x^2) \cdot x^{1/2} dx$ $= \int_0^1 (x^{1/2} - 2x^{3/2} + x^{5/2}) dx$ $= \left[\frac{x^{3/2}}{3/2} - \frac{2x^{5/2}}{5/2} + \frac{x^{7/2}}{7/2} \right]_0^1$ $= \left[\frac{2}{3} - \frac{4}{5} + \frac{2}{7} \right] - [0 - 0 + 0]$ $= \frac{16}{105}$		$\frac{du}{dx} = -1 \Rightarrow du = -dx$ 1 for showing independent of variable 1 for rearranging integral 1 for answer

Qn	Solutions	Marks	Comments & Criteria
13	(a) To prove $RT \perp MB$ $RANS$ is a cyclic quadrilateral as opposite $\angle$'s supplementing $\angle RAS + \angle RNS = 90^\circ + 90^\circ = 180^\circ$ (1) $\angle ASN = \angle ANS$ (angles at circumference on same segment equal) (2) $\angle ANS = \angle ABM$ (angles at circumference standing on same arc AN are equal) (3) $\angle RSA = \angle TSB$ (vertically opposite angles equal) $\therefore \triangle RAS \sim \triangle STB$ (Equiangular) $\therefore \angle STB = \angle RAS = 90^\circ$ (Corresponding angles in similar triangles are equal) $\therefore RT$ is perpendicular to MB. $\therefore$ proven		

Qn	Solutions	Marks	Comments & Criteria
14	(a)  Triangle is equilateral with side length $2y$ Area of $\Delta = \frac{1}{2}ab \sin c$ $= \frac{1}{2} \times 2y \times 2y \times \sin 60$ $= 2y^2 \times \frac{\sqrt{3}}{2}$ $= \sqrt{3} y^2$ $V = \int_{-4}^4 \sqrt{3} y^2 dx$ $= 2\sqrt{3} \int_0^4 9 \left(1 - \frac{x^2}{16}\right) dx$ $= 18\sqrt{3} \int_0^4 \left(1 - \frac{x^2}{16}\right) dx$ $= 18\sqrt{3} \left[x - \frac{x^3}{48} \right]_0^4$ $= 18\sqrt{3} \left[\left(4 - \frac{64}{48}\right) - (0 - 0) \right]$ $= 18\sqrt{3} \times \frac{8}{3}$ $= 48\sqrt{3} \text{ u}^3$	—	Area Δ rearranging integral Answer

Qn	Solutions	Marks	Comments & Criteria
14 (b) (i)			Asymptotes shape
	(ii) $\frac{x^2 - 6}{x - 2} = mx + b + \frac{a}{x - 2}$ $\begin{array}{r} x + 2 \\ x - 2 \overline{) x^2 - 0x - 6} \\ \underline{x^2 - 2x} \\ 2x - 6 \\ \underline{2x - 4} \\ -2 \end{array}$ $\therefore \frac{x^2 - 6}{x - 2} = x + 2 - \frac{2}{x - 2}$ $\therefore$ Eqn of ℓ: $y = x + 2$		proper correct soln.
14 (c) (i)	$(\cos \theta + i \sin \theta)^3$ $= \cos^3 \theta + 3 \cos^2 \theta \cdot i \sin \theta + 3 \cos \theta \cdot i^2 \sin^2 \theta + i^3 \sin^3 \theta$ $= \cos^3 \theta - 3 \cos \theta \sin^2 \theta + i (3 \cos^2 \theta \sin \theta - \sin^3 \theta)$ $= \cos^3 \theta - 3 \cos \theta (1 - \cos^2 \theta) + i (3 \cos^2 \theta \sin \theta - \sin^3 \theta)$ $= \cos^3 \theta - 3 \cos \theta + 3 \cos^3 \theta + i (3 \cos^2 \theta \sin \theta - \sin^3 \theta)$ $= 4 \cos^3 \theta - 3 \cos \theta + i (3 \cos^2 \theta \sin \theta - \sin^3 \theta)$		

Qn	Solutions	Marks	Comments & Criteria
	$z = cis \theta$ $z^3 = (cis \theta)^3$ $= 1^3 cis 3\theta$ $z^3 = \cos 3\theta + i \sin 3\theta$ Equating the parts $\cos 3\theta = 4\cos^3 \theta - 3\cos \theta$ cii) $8x^3 - 6x - 1 = 0$ Let $x = \cos \theta$ i.e. $8\cos^3 \theta - 6\cos \theta - 1 = 0$ $2(4\cos^3 \theta - 3\cos \theta) - 1 = 0$ $2 \cos 3\theta - 1 = 0$ $2 \times \frac{1}{2} - 1 = 0$ where $\cos 3\theta = \frac{1}{2}$ $\therefore \cos \theta = x$ is a soln. ciii) $\cos 3\theta = \frac{1}{2}$ $3\theta = \cos^{-1}(\frac{1}{2})$ $3\theta = \frac{\pi}{3} + 2n\pi$ for $n = 0, \pm 1, \pm 2$ etc $= \frac{\pi + 6n\pi}{3}$ $3\theta = \frac{\pi(6n+1)}{3}$ $\theta = \frac{\pi(6n+1)}{9}$ $\theta = \frac{\pi}{9}, \frac{5\pi}{9}, \frac{7\pi}{9}$ $\therefore$ Solns are $\cos \frac{\pi}{9}, \cos(\frac{5\pi}{9}), \cos(\frac{7\pi}{9})$		1 for showing 1 1

Qn	Solutions	Marks	Comments & Criteria
	(iv) $\cos \frac{5\pi}{9} = -\cos \frac{4\pi}{9}$ $\cos \frac{7\pi}{9} = -\cos \frac{2\pi}{9}$ $\therefore \cos \frac{\pi}{9} \cdot \cos \frac{5\pi}{9} \cdot \cos \frac{7\pi}{9}$ $= \cos \frac{\pi}{9} (-\cos \frac{4\pi}{9}) (-\cos \frac{2\pi}{9})$ $= \cos \frac{\pi}{9} \cos \frac{2\pi}{9} \cos \frac{4\pi}{9}$ $= \text{Product of roots.}$ $\angle 38 = -(-\frac{1}{8})$ $= \frac{1}{8}$ $\therefore \cos \frac{\pi}{9} \cos \frac{2\pi}{9} \cos \frac{4\pi}{9} = \frac{1}{8}$	1	for equating 1 value
15 (a, i)	$x^2 - y^2 = 9$ $\frac{x^2}{9} - \frac{y^2}{9} = 1$ $b^2 = a^2(e^2 - 1)$ $9 = 9(e^2 - 1)$ $1 = e^2 - 1$ $e^2 = 2$ $e = \sqrt{2} \therefore \text{A rectangular hyperbola}$	or As $a^2 = b^2$ i.e. $a = b$ $\therefore$ A rectangular hyperbola $\therefore e = \sqrt{2}$	1
(ii)	$a = 3, b = 3$ Foci : $(\pm ae, 0)$ $= (\pm 3\sqrt{2}, 0)$ Directrices : $x = \pm \frac{a}{e}$ $= \pm \frac{3}{\sqrt{2}} = \pm \frac{3\sqrt{2}}{2}$ Asymptotes $y = \pm x$ (as $e = \sqrt{2}$)		1

or $y = \pm \frac{bx}{a}$

$= \pm \frac{3x}{3}$

$= \pm x$

Qn	Solutions	Marks	Comments & Criteria
15	(iii) $x^2 - y^2 = 9$ $2x - 2y \frac{dy}{dx} = 0$ $\frac{dy}{dx} = \frac{2x}{2y}$ $= \frac{x}{y}$ At P, $m_T = \frac{x_1}{y_1}$ Eqn $y - y_1 = m(x - x_1)$ $y - y_1 = \frac{x_1}{y_1}(x - x_1)$ $y_1 y - y_1^2 = x_1 x - x_1^2$ $yy_1 = xx_1 - (x_1^2 - y_1^2)$ $yy_1 = xx_1 - 9$ (iv) At Q $y = x$ $\therefore xy_1 = xx_1 - 9$ $x = \frac{9}{x_1 - y_1}$ $\therefore y = \frac{9}{x_1 - y_1}$ Area $\triangle QOR = \frac{1}{2}bh = \frac{1}{2} \cdot OQ \cdot OR$ $OQ^2 = \left(\frac{9}{x_1 - y_1}\right)^2 + \left(\frac{9}{x_1 - y_1}\right)^2$ $= \frac{162}{(x_1 - y_1)^2}$ $OQ = \frac{9\sqrt{2}}{x_1 - y_1}$	as $x^2 - y^2 = 9$ i.e. $x_1^2 - y_1^2 = 9$ At R $y = -x$ $-xy_1 = xx_1 - 9$ $x = \frac{9}{x_1 + y_1}$ $\therefore y = \frac{-9}{x_1 + y_1}$ $OR^2 = \left(\frac{9}{x_1 + y_1}\right)^2 + \left(\frac{-9}{x_1 + y_1}\right)^2$ $= \frac{162}{(x_1 + y_1)^2}$ $OR = \frac{9\sqrt{2}}{(x_1 + y_1)}$	1 1 1

Qn	Solutions	Marks	Comments & Criteria
	$\therefore \text{Area } \triangle QOA$ $= \frac{1}{2} \times \frac{9\sqrt{x}}{x_1 - y_1} \times \frac{9\sqrt{x}}{x_1 + y_1}$ $= \frac{81}{x_1^2 - y_1^2}$ $= \frac{81}{9}$ $= 9u^2 \quad \therefore \text{a constant value.}$ (b) $y = x^3 - px + q$ (i) $y' = 3x^2 - p$ $= 0 \text{ when } 3x^2 = p$ $x = \pm \sqrt{\frac{p}{3}}$ $\therefore 2 \text{ turning pts when } p > 0$ (ii) At $x = \sqrt{\frac{p}{3}}$ $y = \left(\sqrt{\frac{p}{3}}\right)^3 - p\sqrt{\frac{p}{3}} + q$ $= \frac{p}{3}\sqrt{\frac{p}{3}} - p\sqrt{\frac{p}{3}} + q$ $= -\frac{2p}{3}\sqrt{\frac{p}{3}} + q$ $= q - \frac{2p}{3}\sqrt{\frac{p}{3}}$ At $x = -\sqrt{\frac{p}{3}}$ $y = \left(-\sqrt{\frac{p}{3}}\right)^3 + p\sqrt{\frac{p}{3}} + q$ $= -\frac{p}{3}\sqrt{\frac{p}{3}} + p\sqrt{\frac{p}{3}} + q$ $= \frac{2p}{3}\sqrt{\frac{p}{3}} + q$ $= q + \frac{2p}{3}\sqrt{\frac{p}{3}}$ $\therefore \text{Min at } \left(\sqrt{\frac{p}{3}}, q - \frac{2p}{3}\sqrt{\frac{p}{3}}\right)$ $\text{Max at } \left(-\sqrt{\frac{p}{3}}, q + \frac{2p}{3}\sqrt{\frac{p}{3}}\right)$ For 3 intercepts, k lies between max and min $\therefore q - \frac{2p}{3}\sqrt{\frac{p}{3}} < k < q + \frac{2p}{3}\sqrt{\frac{p}{3}} \text{ as req'd}$		

Qn	Solutions	Marks	Comments & Criteria
15	(c) $x^3 - x^2 - 3x + 5 = 0$ Roots α, β, γ Sum of roots $\alpha + \beta + \gamma = \frac{-(-1)}{1}$ $= 1$ Roots of eqn $2\alpha + \beta + \gamma, \alpha + 2\beta + \gamma, \alpha + \beta + 2\gamma$ Now $2\alpha + \beta + \gamma = \alpha + \beta + \gamma + \alpha$ $= 1 + \alpha$ $\therefore$ Roots of new eqn are $1 + \alpha, 1 + \beta, 1 + \gamma$ $x = 1 + \alpha$ $\Rightarrow \alpha = x - 1$ etc $\therefore$ Eqn $(x-1)^3 - (x-1)^2 - 3(x-1) + 5 = 0$ $x^3 - 3x^2 + 3x - 1 - x^2 + 2x - 1 - 3x + 3 + 5 = 0$ $x^3 - 4x^2 + 2x + 6 = 0$		
16	(a) $T_n = T_{n-1} + d$ (i) $\frac{1}{\sqrt{T_{n-1}} + \sqrt{T_n}} \times \frac{\sqrt{T_{n-1}} - \sqrt{T_n}}{\sqrt{T_{n-1}} - \sqrt{T_n}}$ $= \frac{\sqrt{T_{n-1}} - \sqrt{T_n}}{T_{n-1} - T_n}$ $= \frac{\sqrt{T_{n-1}} - \sqrt{T_n}}{T_{n-1} - (T_{n-1} + d)}$ $= \frac{\sqrt{T_{n-1}} - \sqrt{T_n}}{-d}$		Rationalising Denom.

$$= \frac{\sqrt{T_n} - \sqrt{T_{n-1}}}{d} \text{ as req'd}$$

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Qn	Solutions	Marks	Comments & Criteria
16	(a)(ii) From (i) $\frac{1}{\sqrt{T_1} + \sqrt{T_2}} = \frac{\sqrt{T_2} - \sqrt{T_1}}{d}$ etc $\therefore \frac{1}{\sqrt{T_1} + \sqrt{T_2}} + \frac{1}{\sqrt{T_2} + \sqrt{T_3}} + \dots + \frac{1}{\sqrt{T_{n-1}} + \sqrt{T_n}}$ $= \frac{\sqrt{T_2} - \sqrt{T_1}}{d} + \frac{\sqrt{T_3} - \sqrt{T_2}}{d} + \dots + \frac{\sqrt{T_n} - \sqrt{T_{n-1}}}{d}$ $= \frac{1}{d} [\sqrt{T_2} - \sqrt{T_1} + \sqrt{T_3} - \sqrt{T_2} + \dots + \sqrt{T_n} - \sqrt{T_{n-1}}]$ etc $= \frac{\sqrt{T_n} - \sqrt{T_1}}{d} \times \frac{\sqrt{T_n} + \sqrt{T_1}}{\sqrt{T_n} + \sqrt{T_1}}$ $= \frac{T_n - T_1}{d(\sqrt{T_n} + \sqrt{T_1})}$ Now $T_n = T_1 + (n-1)d$ i.e. $u_n = a + (n-1)d$ $\therefore T_n - T_1 = (n-1)d$ $= \frac{(n-1)d}{d(\sqrt{T_n} + \sqrt{T_1})}$ $= \frac{n-1}{\sqrt{T_n} + \sqrt{T_1}}$ $= \frac{n-1}{\sqrt{T_1} + \sqrt{T_n}} \text{ as req'd}$		1 1

Qn	Solutions	Marks	Comments & Criteria
16	(b) $3x^2 + 4y^2 = 48$ $3x^2 - y^2 = 3$ $\frac{x^2}{16} + \frac{y^2}{12} = 1$ $x^2 - \frac{y^2}{3} = 1$ (i) Foci: Hyperbola Ellipse $b^2 = a^2(e^2 - 1)$ $b^2 = a^2(1 - e^2)$ $3 = 1(e^2 - 1)$ $12 = 16(1 - e^2)$ $3 = e^2 - 1$ $12 = 16 - 16e^2$ $e^2 = 4$ $-4 = -16e^2$ $e = 2$ $e^2 = \frac{1}{4}$ $\therefore ae = 2$ $e = \frac{1}{2}$ Foci: $(\pm ae, 0)$ $\therefore$ Foci $(\pm ae, 0)$ $= (\pm 2, 0) *$ i.e. $(\pm 2, 0) *$ $\therefore$ Same foci (ii) $(4\cos\theta, 2\sqrt{3}\sin\theta)$ satisfies $3x^2 - y^2 = 3$ $3(16\cos^2\theta) - 12\sin^2\theta = 3$ $16\cos^2\theta - 4\sin^2\theta = 1$ $16\cos^2\theta - 4(1 - \cos^2\theta) = 1$ $16\cos^2\theta - 4 + 4\cos^2\theta = 1$ $20\cos^2\theta = 5$ $\cos^2\theta = \frac{1}{4}$ $\cos\theta = \pm \frac{1}{2}$ $\therefore \theta = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}$		

At $\theta = \frac{\pi}{3}$ $(4 \times \frac{1}{2}, 2\sqrt{3} \times \frac{\sqrt{3}}{2}) = (2, 3)$

By symmetry: Pts of intersection $(2, 3)$ $(2, -3)$
 $(-2, 3)$ $(-2, -3)$

Qn	Solutions	Marks	Comments & Criteria
Q16	(b)(iii) Show intersect at right angles i.e. show tangents intersect at right angles $m_1 m_2 = -1$ $3x^2 + 4y^2 = 48$ $6x + 8y \frac{dy}{dx} = 0$ $\frac{dy}{dx} = -\frac{6x}{8y}$ $= -\frac{3x}{4y} \quad (m_1)$ $3x^2 - y^2 = 3$ $6x - 2y \frac{dy}{dx} = 0$ $\frac{dy}{dx} = \frac{6x}{2y}$ $= \frac{3x}{y}$ At (2, 3) $m_1 = -\frac{6}{12} = -\frac{1}{2}$ $m_2 = \frac{6}{3} = 2$ } $m_1 m_2 = -1$ At (2, -3) $m_1 = \frac{-6}{-12} = \frac{1}{2}$ $m_2 = \frac{6}{-3} = -2$ } $m_1 m_2 = -1$ At (-2, 3) $m_1 = \frac{6}{12} = \frac{1}{2}$ $m_2 = \frac{-6}{3} = -2$ } $m_1 m_2 = -1$ At (-2, -3) $m_1 = \frac{-6}{-12} = \frac{1}{2}$ $m_2 = \frac{6}{-3} = -2$ } $m_1 m_2 = -1$		

$\therefore$ Intersect at right angles

Qn	Solutions	Marks	Comments & Criteria
16	(c) (iii) $n I_n = (n-1) I_{n-2}$ i.e. $I_n = \frac{(n-1)}{n} I_{n-2}$ $I_5 = \frac{4}{5} \times I_3$ $I_3 = \frac{2}{3} I_1$ $I_1 = \int_0^1 (1-x^2)^{-\frac{1}{2}} dx$ $= \int_0^1 (1-x^2)^0 dx$ $= \int_0^1 1 \cdot dx$ $= [x]_0^1$ $= 1$ $\therefore I_3 = \frac{2}{3}$ $I_5 = \frac{4}{5} \times \frac{2}{3}$ $= \frac{8}{15}$		1